Practical assignment AZA 2025/26 – 20 points, deadline to submit 5/12/2025, 5 PM

1. **Design and implement an algorithm to solve a problem for scheduling with deadlines (Algorithm 4.4).**

Scheduling with Deadlines

Problem: Determine the schedule with maximum total profit given that each job has a profit that will be obtained only if the job is scheduled by its deadline.

Inputs: n, the number of jobs, and array of integers *deadline*, indexed from 1 to n, where *deadline*[i] is the deadline for the ith job. The array has been sorted in nonincreasing order according to the profits associated with the jobs.

Outputs: an optimal sequence J for the jobs.

```
void schedule (int n,
               const int deadline[],
               sequence_of_integer& J)
{
  index i;
  sequence_of_integer K;

  J = [1];
  for (i = 2; i <= n; i++){
     K = J with i added according to nondecreasing values of deadline[i];
     if (K is feasible)
        J = K;
  }
}
```

Table 1.1. Consider the following jobs, deadlines, and profits, and use the scheduling with deadlines algorithm to maximize the total profit. Show the optimal sequence of jobs with max profit on the screen.

Job	Deadline	Profit
1	2	40
2	4	15
3	3	60
4	2	20
5	3	10
6	1	45
7	1	55

2. **Consider the procedure schedule in the Scheduling with Deadlines algorithm (Algorithm 4.4). Let d be the maximum of the deadlines for n jobs. Modify the procedure so that it adds a job as late as possible to the schedule being built, but no later than its deadline. Do this by initializing d+1 disjoint sets, containing the integers 0, 1, ..., d. Let small(S) be the smallest member of set S. When a job is scheduled, find the set S containing the minimum of its deadline and n. If small(S) = 0, reject the job. Otherwise, schedule it at time small(S), and merge S with the set containing small(S)−1. Assuming we use Disjoint Set Data**

Structure III in Appendix C, show that this version is θ(nlgm), where m is the minimum of d and n.

3. Huffman compression

Your task is to design and implement simple Huffman compression and decompression. Huffman coding is a lossless data compression algorithm. It assigns codes of varying lengths to characters, where more frequent characters receive shorter codes.

3.1Compression

Your task is to compress files using Huffman coding. The algorithm will be performed on the entire ASCII table, meaning your map will contain every character represented as a numeric key.

Huffman compression works best on larger files. For smaller files, the compressed version may be larger than the original because the entire frequency table and code must be written to the file.

Guidance

Header size → number of bits present in freq map

Pagination of code which len <8

Binary read/write

Use of std::bitset<8>

Example text comparison:

```
md5sum test.txt test_comp.txt
   d57c51344ef4d66c796e14fcf4b185ad  test.txt  14kb
   0e2c98fe11a5b342e52224ec95a51c68  test_comp.txt  6kb
```

3.2 Decompression

After you have successfully completed the compression of files, you will need to create and implement a decompression mechanism in order to verify and confirm that the compression process was flawless and that no data was lost during the compression operation.

Verification

Your decompressed file md5sum should be same as the original.

```
md5sum test.txt test_uncomp.txt
   d57c51344ef4d66c796e14fcf4b185ad  test.txt
   d57c51344ef4d66c796e14fcf4b185ad  test_uncomp.txt
```

Algorithm analysis

Whats the T(n)

Whats the O(T(n))

What would you do to improve the complexity of your solution

What you should hand in

Single ZIP file via AIS system. Name convention for the ZIP file:

FirstName_ LastName _ AZA_practical.ZIP

ZIP file should include four CPP files and a single PDF document with your complexity analysis. No other format, such as Word or Open Office, should be included.
Note 1: Any use of an AI assistant must be referenced; otherwise, it will be considered plagiarism.
Note 2: Late submission, missing files, or ZIP files that cannot be opened and expanded will cost you zero marks for the assignment.
Note 3: It is individual work. Two or more identical submissions will be investigated as plagiarism.
Submission Deadline: 5th December 2025 at 5 PM

Marking table

Huffman solution proposal	1b
Huffman compression	3b
Huffman decompression	3b
Huffman complexity analysis	3b
Scheduling with deadline algorithm implementation	3b
Use of Disjoint Set Data Structure III	3b
Complexity analysis of both approaches	3b
Correct output on a screen based on input from the table	1b